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Galinstan-based flexible microfluidic device for wireless human-sensor applications

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ABSTRACT

Liquid-metal-based wearable technologies are one of the promising ways to realize soft-skin flexible electronics. Here, we report a flexible microfluidic device that injects Galinstan liquid-metal alloy through the microchannels of a polydimethylsiloxane (PDMS) substrate, targeting an inductor-capacitor (LC) resonant circuit for wireless power transmission and sensing. The surface inside PDMS microchannels is chemically modified using sulfuric acid (H_2SO_4) solution, making it easier for the liquid metal to flow through the microfluidic channel. Electromechanical characteristics of the capacitive pressure sensors were evaluated by loading and unloading the external pressure continuously in the range of 0–200 mmHg for 300 cycles. After that, the characterization of Galinstan-based flexible microfluidic sensors consisting of the LC circuit conducted in a strain range of up to 30 %. Finally, monitoring of various human motion (e.g., wrist flexion and finger motion) was demonstrated using a wireless sensing platform with an external antenna. Electrical and mechanical performances confirm that the Galinstan-based sensors are reliable, reproducible, repeatable, and flexible with a sensitivity of 5 kHz/mmHg. The proposed flexible microfluidic device has the potential to become an exceptionally reliable candidate for wireless human-machine applications.

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1. Introduction

The development of flexible electronics has received significant attention in human-machine applications (e.g., wearable electronics, soft-skin robotics, and healthcare systems) because of their high degree of deformability and conformability [1–7]. Over the years, there have been numerous flexible electronics technologies proposed, including polyimide and nickel-chrome based strain-gauge sensors used for detecting resistance changes [8], implanting carbon or silver nanowires into elastomer as pressure sensing elements [9], and spiral-shaped silicon membrane-based resistance measurement systems using the piezoelectric effect for full-body pressure and temperature mapping [10]. However, the properties of conventional flexible electronics (i.e., hardness, brittleness, low flexibility, and rigidity) arising from their solid-state materials make their operation difficult and unreliable on uneven, deformable, and soft surfaces (e.g., human skin). Therefore, many research efforts in the recent years have driven the concept of liquid metal-based flexible electronics dedicated to providing the novel properties (i.e., stretchability, high flexibility, and compliancy).

The liquid metal-based flexible electronics embed conductive liquids into polymer microchannels to form sensors. These sensors can undergo extreme deformation while retaining a complete circuit due to good healing capability of the liquid metal [1112]. Among several liquid metal alloys, Galinstan (a eutectic composition of gallium-68 %, indium-22 %, and tin-10 %) has been widely explored in the flexible electronics due to its electromechanical properties. It has low viscosity and high surface tension as a 'liquid' while, it has high boiling point and high conductivity as a 'metal' like mercury. However, its low toxicity and ultralow vapor pressure render it much safer than mercury. Galinstan forms an oxide layer when exposed to air, resulting in the formation of non-equilibrium structures [13]. This layer causes the liquid metal to "wet" and adhere to the surface, limiting its movement inside the polymer microchannel. Although the thin oxide layer (0.7–3 nm) may not have a significant influence on bulk liquid metal conductors [14], it substantially impacts the fabrication of micro-level devices. Recently, efforts have been made to avoid the oxidation

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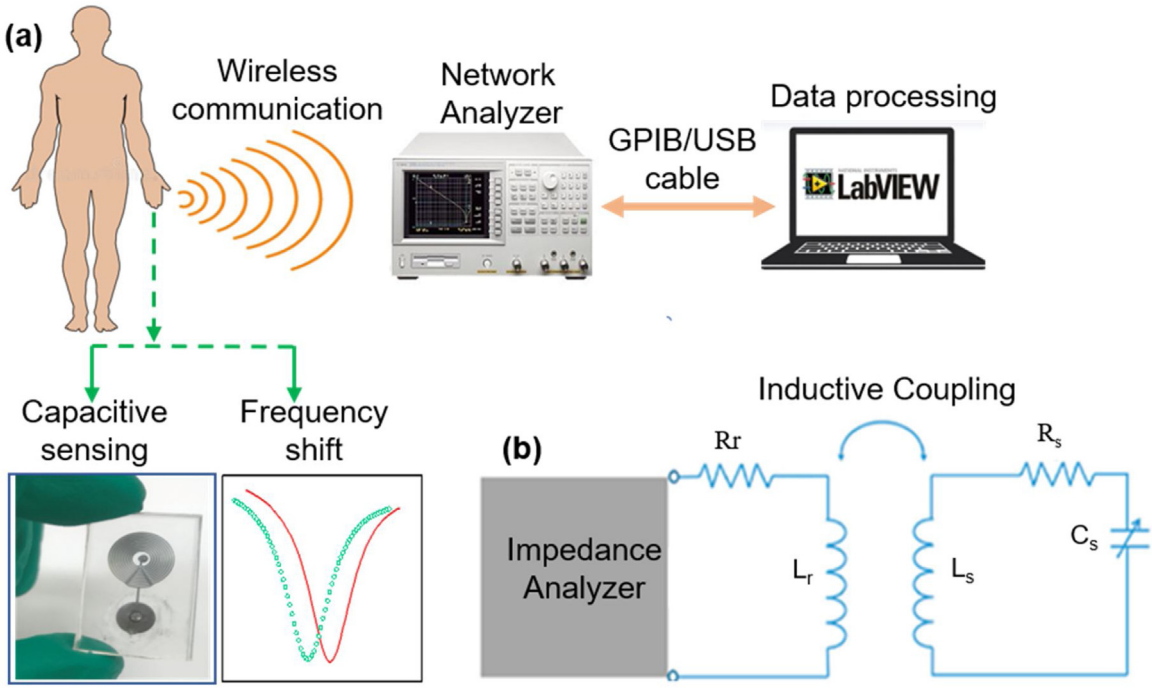

Fig. 1. (a) Schematic of flexible microfluidic device-based wireless capacitive sensor. (b) equivalent circuit of inductively coupled L–C resonator: R_s, L_s and C_s is the resistor, inductor, and variable capacitor of the sensor, respectively. R_r and L_r are the resistor and inductor of the receiver connected to the network analyzer.

of Galinstan by various surface modification techniques [1516] (e.g., physical, chemical, and deoxidization). Therefore, a better environment is needed for implanting Galinstan into the polymer microchannels during the fabrication of liquid metal-based flexible electronics.

In the recent years, various fabrication techniques (e.g., liquid metal printing, selective plating, microfluidic approach, direct writing, and spray-deposition) have been used to realize liquid metal-based flexible electronics [17–20]. Among these, the microfluidic approach is widely employed thanks to its attractive advantages (i.e., compact size, rapid processing, and low energy consumption). The existing microfluidic approach [21–24] has made effective use of highly elastic and biocompatible polymer materials, such as polydimethylsiloxane (PDMS), to define the microchannels less than 100 μm [25]. Thus, several microfluidic devices based on polymer materials have been realized for stretchable biosensor applications [26–29]. While the proposed methods have made it easier to inject liquid metal into a microfluidic channel compared to conventional methods, there are still limitations in the multi-layered microchannel design. These are due to the liquid metal must travel long distances through the channels without sticking to the surfaces and connect all the components in the flexible electronics. To the best of our knowledge, we still believe that there is a need for further development in effective design and fabrication process. This motivated us to conduct further research on liquid metal-based flexible electronics.

This paper demonstrates a liquid metal-based microfluidic device for wireless sensor applications. The proposed wireless sensor is designed by applying an inductor-capacitor (LC) resonant circuit [3031]. Each component in the LC configuration is electrically connected using the Galinstan on a lithographically defined PDMS microchannel. The flexibility and stretchability of the electronic devices have increased significantly by employing the liquid metal instead of conventional solid metal materials. The inner walls of PDMS are treated with H_2SO_4, allowing the Galinstan to move easily in microchannels. The fabricated flexible sensor makes the most of multilayer microchannel, employing a 10-turn inductor coil connected to each capacitor plate. After preliminary experiment, important characteristics (i.e., reliability, flexibility, repeatability, and reproducibility) of the sensors have been successfully conformed by evaluating key parameters such as pressure loading and unloading test, strain test, and human-motion monitoring.

2. Experimental section

2.1. Principle of wireless capacitive sensors

In recent years, wireless human-signal monitoring has been implemented using flexible electronics with capacitive, piezoresistive, triboelectric, and piezoelectric sensing mechanisms [32–36]. Among those, capacitive sensing presents low-pressure detection limits, short response time, high working stability, and low power consumption features. Fig.1 shows the schematic illustration of proposed wireless capacitive sensor. It employs the LC resonant circuit with parallel plate variable capacitor as the sensing element and a circular spiral coil as the sensor antenna for wireless communication based on inductive coupling. The circular spiral coils are reliable for wearable devices and implantation under the skin due to their robustness against lateral coil misalignment and enhance the coil performance in air and other mediums [37]. On the other hand, the design of circular inductor coils is limited by many factors such as shape, size, separation distance number of turns. In order to optimize the sensor antenna, an LC circuit with 5-turn inductor coil having a resonant frequency of 495 MHz was designed and fabricated as shown in the supporting file Fig. S1. However, the network analyzer (model 4395A) with the frequency range of 100 kHz-500 MHz could not read the resonant frequency of the sensor. This may be due to: 1) the resonant frequency does not lie in the range of the network analyzer or, 2) the power generated by sensor antenna is not sufficient for wireless communication. On further trial, the sensor antenna with 10-turn inductor coils were tested and found to be ideal for our measurement system. The LC resonant circuit's design and specifications are provided in the supplementary document under the design section (see Fig. S2). Here, the capacitive

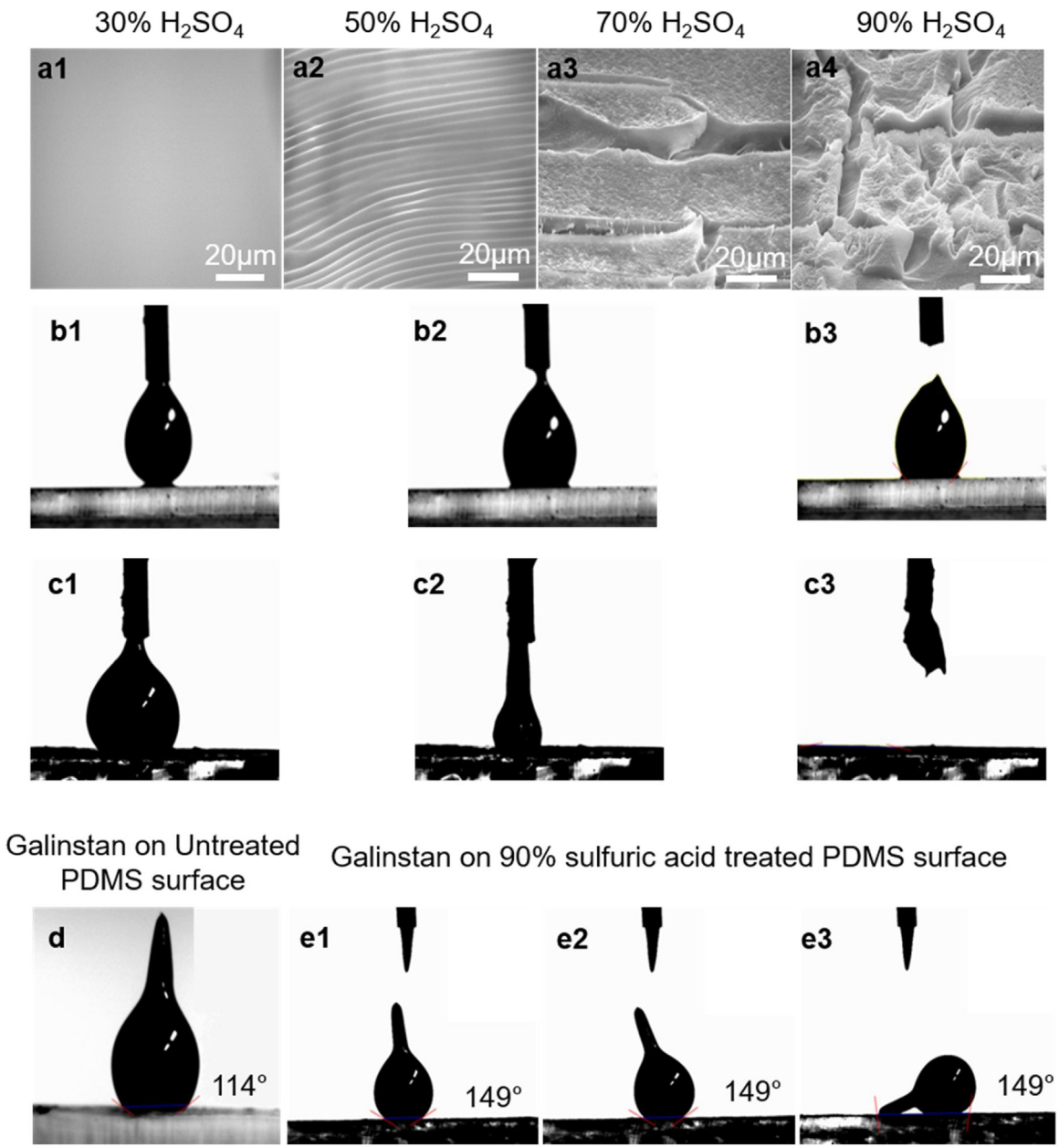

Fig. 2. (a1–a4) SEM image of H_2SO_4-treated PDMS microchannel for different concentrations (30 to 90 %) for 1 min, respectively. (b1–b3) Galinstan sticks to the untreated PDMS surface while being sucked up by the syringe. (c1–c3) Galinstan does not stick (no residue) to the H_2SO_4-treated PDMS surface while being sucked up by the syringe. (d and e1–e3) behavior and contact angle of Galinstan on untreated and 90 % H_2SO_4-treated PDMS surface as a function of time.

sensor and the internal antenna is built on the microfluidic device with Galinstan as the conductive metal. A PDMS dielectric layer is sandwiched between the capacitor plates, acting as a deforming layer under external pressure. When uniform pressure is applied on the capacitor plate it deforms the dielectric layer and reduces the resonant frequency of the LC resonator by changing the coupling capacitance of the sensor. The change in resonant frequency (F) of the sensor is continuously monitored by the network analyzer using a wireless readout method, expressed as the function of inductance (L_s) and capacitance (C_s) given by:

$$\omega_0 = 2\pi f_0 = \frac{1}{\sqrt{L_s C_s}} \quad (1)$$

The readout method enables wireless operation [38–40] via inductive coupling, which renders the microfluidic device battery-free, owing to wireless powering. In addition, the working distance between the Galinstan-made sensor antenna and the external readout antenna is optimized for wireless data and power communications.

2.2. Chemical modification of the PDMS microchannel

Galinstan normally forms a thin-oxide layer when it exposed to air, causing the liquid metal to stick to surfaces. This oxidized layer interrupts the flow of liquid metal in the PDMS microchannel. Thus, chemical techniques employing HCl and H_2SO_4 have often been employed to efficiently remove the oxide layer formed [1516]. However, the HCl vapor almost pass the PDMS layer and have harmful effects to the human skin. Therefore, instead of reducing the Galinstan using HCl, a method of chemically modifying the inner wall of the PDMS micro channels was investigated. In order to enhance the flow of liquid metal, a lyophobic surface is highly

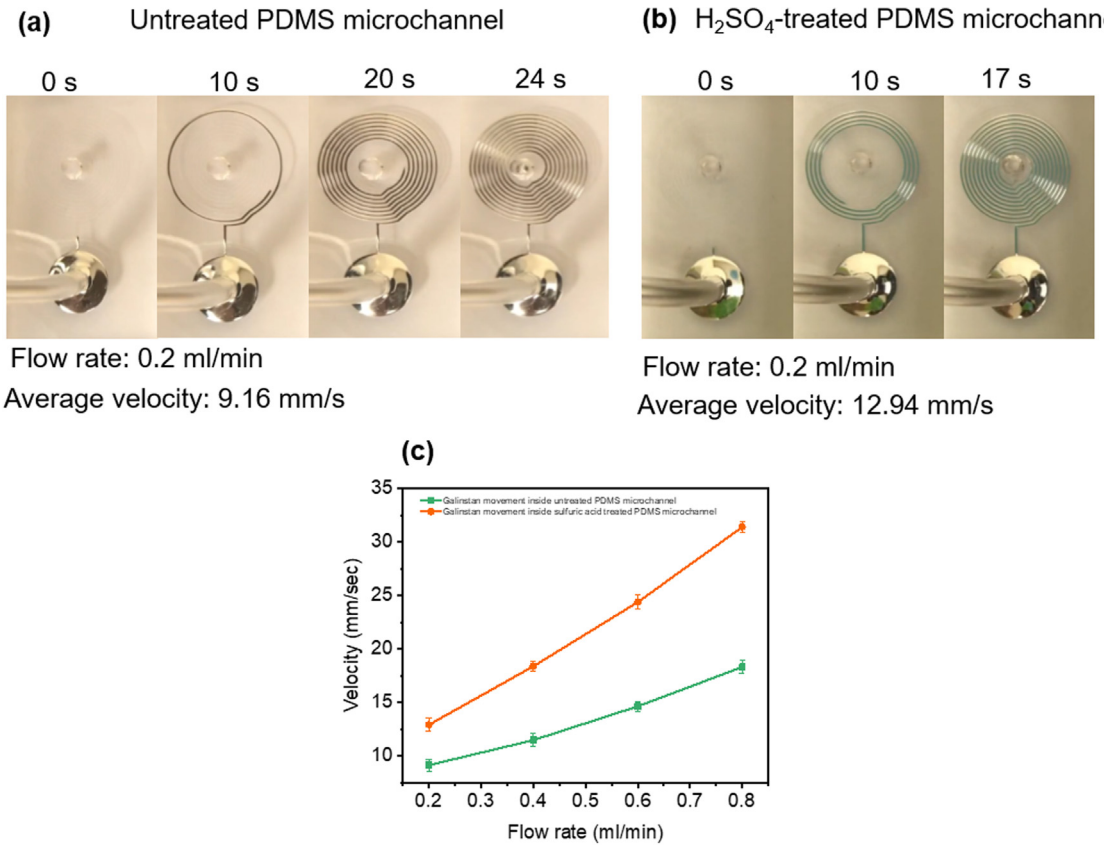

Fig. 3. Movement of Galinstan inside the untreated and H_2SO_4-treated PDMS microchannel of length, 220 mm. (a) Galinstan takes 24 s to fill the untreated PDMS microchannel, (b) 17 s to fill the H_2SO_4-treated PDMS microchannel and (c) comparison of Galinstan's velocity with different flow rates in ml/min (0.2, 0.4, 0.6, 0.8) inside the untreated and H_2SO_4-treated PDMS microchannel.

desired in the PDMS microchannel. Recently, Li et al. reported that the chemically-treated PDMS surface can be directly modified into lyophobic surface which enhances the non-wetting characteristics of Galinstan [16].

To study the reaction between PDMS and H_2SO_4, the PDMS microchannel was treated at a specific concentration of H_2SO_4: water (Part A: Part B) from 30 % (3:7) to 90 % (9:1), for 1 min. Fig. 2(a1- a4) show the scanning electron microscope (SEM) image of the chemically modified PDMS microchannel with different concentration from 30 % to 90 %, respectively. The figure clearly displays that the lower concentration forms micro-bumpy patterns, whereas the nanoscale porous patterns increase with increase of H_2SO_4 concentration. These nanoscale patterns are randomly distributed on the PDMS surface, providing higher lyophobicity and an improved environment for easy flow of Galinstan. Fig. 2(b1-b3) and Fig. 2(c1-c3) shows the comparison of non-wetting characteristics of Galinstan on the untreated and H_2SO_4-treated PDMS surface respectively. It clearly demonstrates that the Galinstan sticks to the untreated PDMS surface when it is sucked up with a syringe, while Galinstan does not stick to the H_2SO_4-treated PDMS surface; it is possible to recover the liquid metal using a syringe, leaving no residue. The non-wetting behavior of Galinstan on the PDMS surface is further demonstrated in the supporting movies, SV1, SV2 and Fig. S3.

The behavior of Galinstan on a 90 % H_2SO_4-treated PDMS surface was evaluated based on the contact angle measurements. The images in Fig. 2(d) and Fig. 2(e1–e3) display the contact angle of Galinstan on an untreated PDMS surface (114°) and a 90 % H_2SO_4-treated PDMS surface (149°), respectively. It indicates that the Galinstan adhere to the pure PDMS with smaller contact angle due to the strong adhesive force between the PDMS surface and the liquid metal, while recovering larger contact angle with H_2SO_4-treated PDMS surface. The larger contact angle is due to strong lyophobic interaction between the nano-particles, influencing the uniformity of the liquid metal on the PDMS surface. Fig. 2(e1–e3) shows that the Galinstan droplet falls as a function of time with constant contact angle, illustrating the non-wetting behavior of Galinstan on the H_2SO_4-treated PDMS surface. Thus, the treatment of H_2SO_4 enhance the reliability of PDMS microchannel to retain the non-wetting characteristics of Galinstan.

In support of the reliability of the H_2SO_4-treated microchannel, the velocity of Galinstan inside untreated and H_2SO_4-treated PDMS microchannels were compared as shown in the supporting movies, SV3-SV10. Here, the 10-turn spiral inductor channel with a length of 220 mm, an inner diameter of 3-mm, an outer diameter of 10.6-mm, and 40-μm thickness was employed to analyze the flow and behavior of Galinstan. The Galinstan was injected using a syringe pump into the PDMS microchannel at the flow rate of 0.2–0.8 mL/min. As a result, the 0.2-ml/min flow rate takes 24 and 17 s to fill the untreated and H_2SO_4-treated PDMS microchannels, with an average velocity of 9.16 mm/s and 12.94 mm/s, respectively, as shown in the Fig. 3(a, b). It demonstrates that the oxide layer reduces the frictional force between the microchannel wall and Galinstan, causing it to stick to the surface, decrease velocity, and take longer to travel in an untreated microchannel. In contrast, the H_2SO_4-treated microchannel improves the non-wetting characteristics, so that the Galinstan does not stick to the surface, increasing the average velocity and taking less time to travel inside the microchannel. As expected, the Galinstan average velocity inside the microchannel increases for the higher flow rate, reaching 20

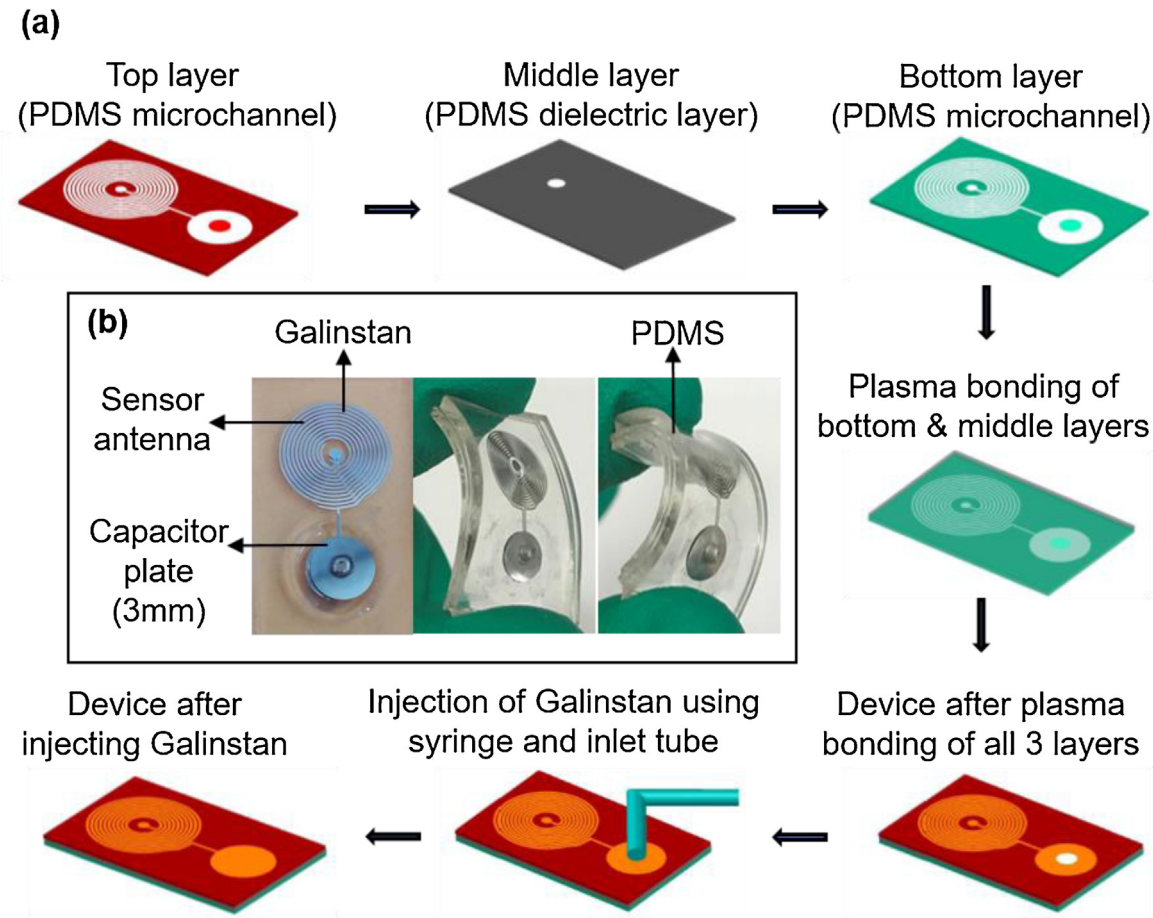

Fig. 4. (a) Schematic of the fabrication of PDMS microfluidic channel-based wireless capacitive sensor using Galinstan as the conductive material and (b) image of the fabricated LC circuit (L: sensor antenna, C: sensing element) flexible microfluidic device.

and 31.5 mm/s for 0.8 mL/min in the untreated and H_2SO_4-treated microchannels, respectively, as displayed in Fig. 3(c). Therefore, all the above experiments support the notion that the chemical modification of PDMS microchannel improve non-wetting characteristics, avoid sticking, and improves electromechanical properties of the sensor.

2.3. Fabrication of microfluidic channel-based capacitive sensor

The wireless flexible microfluidic device is fabricated by implanting Galinstan into a PDMS microchannel using the soft lithography technique. The proposed device has three PDMS layers, as shown in Fig. 4(a), where the top and bottom layer microchannels have identical 10 turns in the inductor coil, 40-μm thick and 200-μm wide. A 3-mm capacitor plate along with inductor forms the LC resonant circuit of the wireless capacitive sensor. The middle layer, having a designed thickness of 0.5 mm, acts as the dielectric medium of the capacitor and has a hole interconnector for the inductor coil. During the fabrication process, a master mold on a silicon wafer for each layer is defined from a SU-8 photoresist using a lithography process. A two-part polymer of a base elastomer (Sylgard 184, Dow Corning Corp., Auburn, MI, USA) and a curing agent is mixed in a ratio of 10-part base elastomer to 1-part curing agent (10:1), poured directly onto the SU-8 patterned silicon wafer and cured on a hot plate at 80 °C for 2 h. After carefully peeling it off, the PDMS microchannels were cut to form top, middle, and bottom layers, as shown in the supplementary Fig. S4. Each layer of the device was then treated with a low-pressure oxygen plasma at 100 W for 60 s and bonded permanently at 80 °C for 1 h to form the microchannel of the wireless capacitive sensor. Next, the PDMS microchannel surface was modified using H_2SO_4 (CAS No: 7664-93-9, Assay: 95–99 %) for smooth and non-sticky flow of Galinstan. Subsequently, Galinstan (Gallium Indium Tin Alloy, Ga: 68.5 % In: 21.5 % Sn: 10 %, Electrical conductivity: 3.46×10^6 S/m, Purity: 99.999 %) was introduced into the microchannel with a syringe, and the inlet and outlet were sealed using cured PDMS. Fig. 4(b) shows the fabricated Galinstan-based flexible microfluidic device for the wireless sensor applications.

3. Result and discussion

3.1. Characterization of liquid-metal-based capacitive sensor

A customized protocol for the characterization of wireless capacitive sensor was implemented using a network analyzer with an external readout antenna of 5-turn inductor coils fixed to it (see Fig. S5). The fabricated capacitive sensor, having a designed resonant frequency of 83 MHz, was glued into a specially made pressure chamber setup connected to a syringe pump and a digital pressure indicator, as displayed in the Fig. 5(a). Then, external pressure, ranging from 0 to 200 mmHg, was applied to the variable capacitor through a 10 mL syringe at 1 mL/min, deforming the dielectric layer and changing the capacitance and resonant frequency of the capacitive sensor. The corresponding resonant frequency was transmitted to the network analyzer through the external antenna placed above the sensor antenna. Fig. 5(b) shows the influence of external pressures on the capacitor, from 0 to 200 mmHg at 40-mmHg interval, reduced the resonant frequency of the sensor from 53.3 MHz to

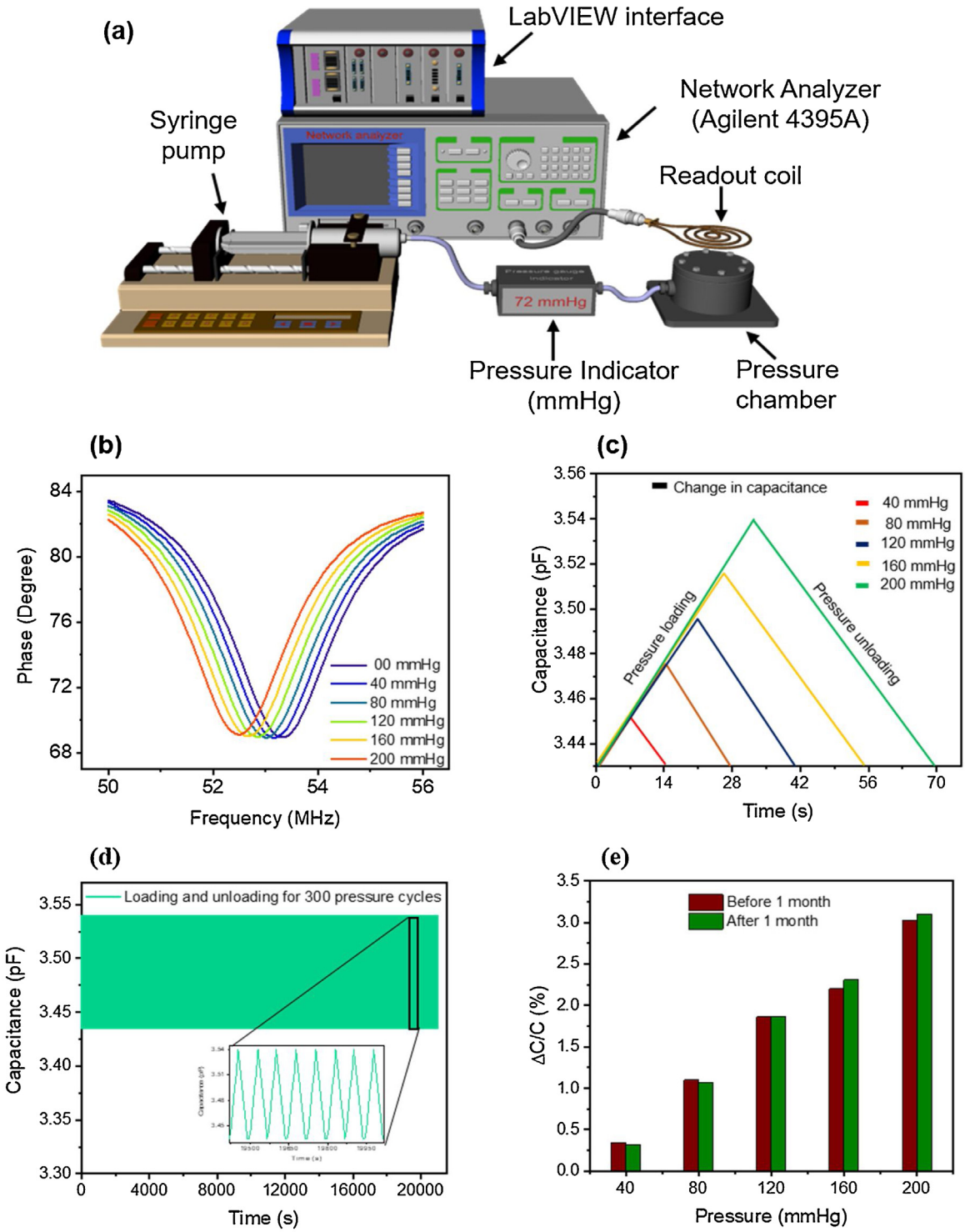

Fig. 5. Characterization of the wireless capacitive sensor. (a) on-bench setup for the study of capacitance sensor under external pressure, (b) phase-dip response of the sensor for the applied pressure at 40-mmHg intervals, (c) loading-unloading behavior of the sensor for various peak pressures at a syringe (10 mL) flow rate of 1 mL/min, (d) change in capacitance as a function of time for continuous loading (0–200 mmHg) and unloading (200–0 mmHg) of 300 pressure cycles, (e) percentage change in capacitance of the sensor before and after 1 month.

52.5 MHz. This is because the external pressure deformed Galinstan made variable capacitor, increasing the capacitance of the sensor for wide pressure. The sensitivity and resonant frequency of the capacitive sensor was 5 kHz/mmHg and 53.3 MHz, respectively.

Thus, the characterization of the capacitive sensor shows that the resonant frequency and capacitance did not match with the theoretical values. As the proposed sensor emphasis more on the fabrication of Galinstan-based flexible microfluidic device, surface modification of PDMS microchannels, wireless capacitive sensing, and human-machine interface, the agreement of theoretical values with the measured values were considered less important. Still, the reasons for the mismatches were to be found. First, there was a variation in thickness of the dielectric-layer of the capacitor during fabrication. Second, there were other dissimilar dimensions (e.g.,

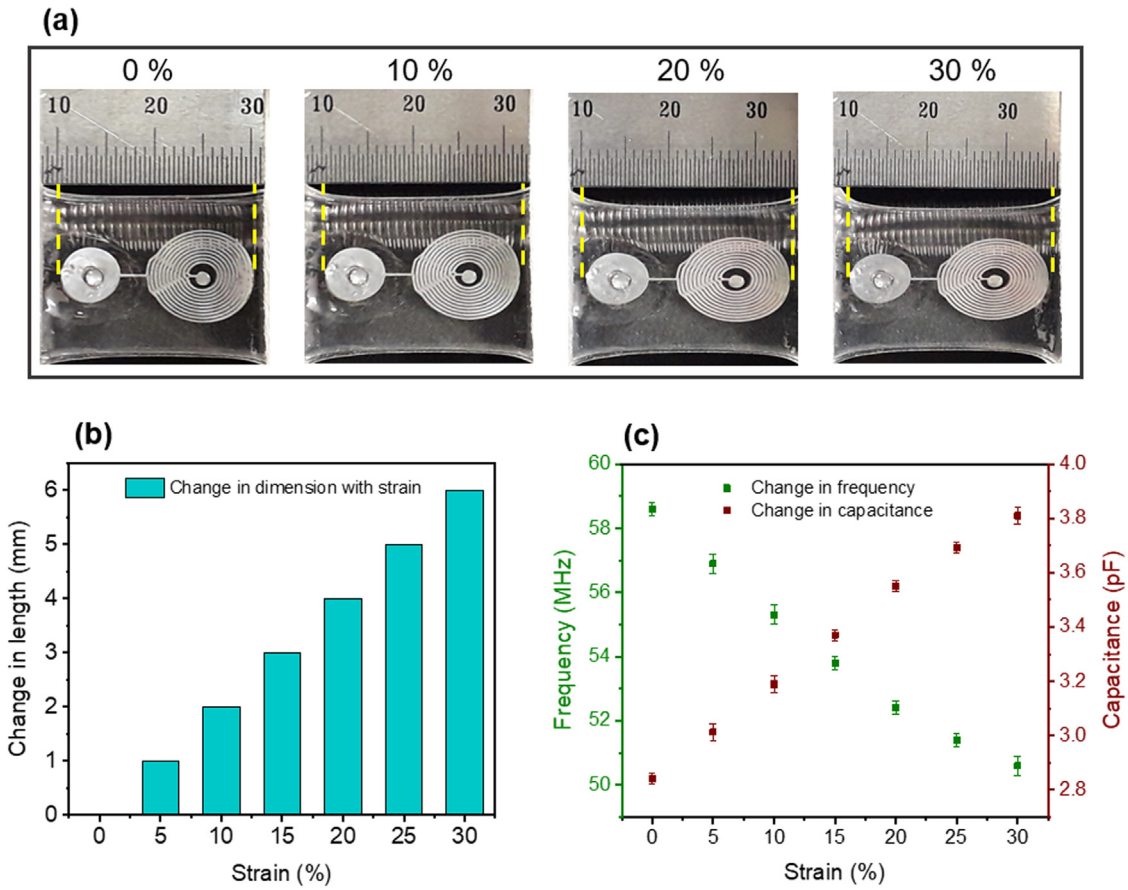

Fig. 6. Electrical and mechanical characteristics of the flexible capacitive sensor under strain (scale bar: 20 mm). (a) image of capacitive sensor stretched from 0% (0 mm) up to 30 % (6 mm), (b) changes in length of the flexible microfluidic device with applied strain and (c) variation of resonant frequency and capacitance with respect to strain.

microchannel dimensions, and thickness of the top and bottom PDMS layers). Nonetheless, the variation in dimensions during the fabrication of microfluidic device could be controlled using process controls, such as microelectromechanical means or by modifying the fabrication method.

One of the main requirements for liquid metal-based capacitive sensor was to maintain their electrical properties during the repeated deformation. Because, the liquid metal remains connected in the normal and highly deformed microchannel, showing good healing capability, while conventional solid metal materials disconnect under high pressure. In order to carefully evaluate the electrical performance, the liquid metal-based capacitive sensor was tested under loading (0–200 mmHg) and unloading (200–0 mmHg) of pressure continuously for 300 cycles. Fig. 5(c) shows the reversible behavior of the sensor under loading-unloading pressure cycles for various pressure peaks, demonstrating capacitance was maximum at 200 mmHg and minimum at 0 mmHg. The sensor also shows stable electrical performance for 300 pressure cycles, repeating and reproducing the effect for all loading-unloading cycles with the minor difference of 2% hysteresis due to low stability strength of the PDMS material, as presented in Fig. 5(d). The enlarged view of the specific time interval, shown in inset of Fig. 5(d), exhibit the reliability and durability of the sensor on repetitive pressure cycles. The response of the sensor was measured again after one month, enhancing the reliability and repeatability of the flexible microfluidic device. Fig. 5(e) shows that the comparison of change in capacitance in percentage. It demonstrates that change in capacitance increased from 0.34 % to 3.03 % (before one month) and 0.32 %–3.14 % (after one month), as the pressure increased from 0–200 mmHg at 40 mmHg interval. The sensor shows good repeatability over a period of one month with minimum difference of 0.02 % and maximum difference 0f 0.11 % in change in capacitance for 0 and 200 mmHg respectively. The corresponding capacitance of the sensor, before and after one month, is shown in supporting file Fig. S6. Thus the characterization of the sensor under loading and unloading of the pressure continuously for 300 cycles and sensor response before and after one month reveals the reliability and repeatability of the Galinstan based capacitive sensor.

3.2. Strain and human-motion monitoring of Galinstan-based flexible microfluidic device

To further explore the potential of Galinstan-based capacitive sensor, the electrical and mechanical properties were observed under various strain tests. The strain test was conducted using a programmatically controlled stepper motor (Autonics A3K-S545) with a single shaft rotating in clockwise and anticlockwise direction. The flexible microfluidic device was fixed on a special arrangement attached to the shaft and stretched linearly at a rate of 1000 μm for each step. The response of the capacitive sensor was analyzed by stretching the device of 20-mm long up to 30 % (26 mm) as shown in Fig. 6(a). The length of the device increased by 6 mm as the strain increased from 0 to 30 %. For every 5 % of strain, there was a 1-mm increase in the length of the device and a total of 6 mm for 30 % strain, as shown in Fig. 6(b). This demonstrates that the capacitive sensor was flexible and could stretch with the surrounding environment. The corresponding response, shown in Fig. 6(c) reveals that the capacitance of the sensor increased as the strain increased from 0 to 30 % (0 mm–6

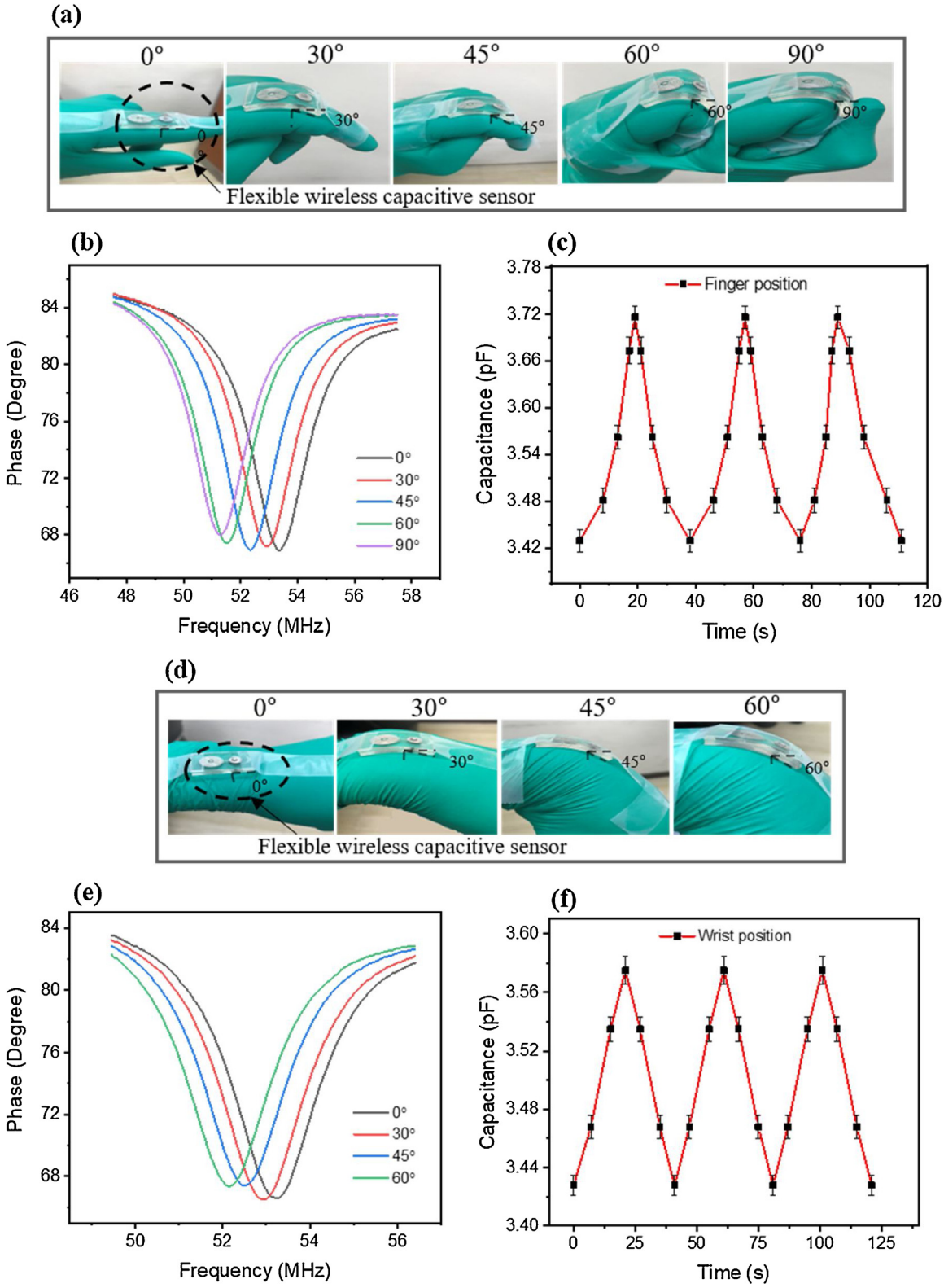

Fig. 7. Flexible wireless capacitive sensor for human-motion monitoring. (a, d) sensor fixed on an index finger and wrist to measure responses at various bending angles, (b, c) changes in resonant frequency and capacitance of the sensor, fixed on the index finger joint, for various bending angles (0–90°) and (e, f) changes in resonant frequency and capacitance of the sensor, fixed on the wrist, for various bending angles (0–60°).

mm) while the resonant frequency decreased, owing to the deformation and changes in dimensions of the capacitor. However, when the applied strain was unloaded, the capacitive sensor regained its initial electrical properties with negligible hysteresis, demonstrating sensor reliability. In addition, the phase response obtained from the sensor in a wireless platform displays that the resonant frequency decreased as the function of strain, shown in supporting file (Fig. S7).

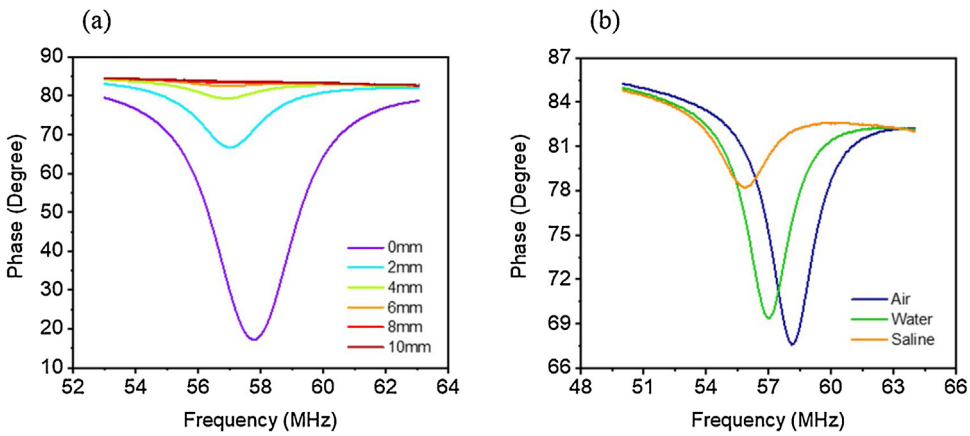

Fig. 8. Wireless communication characterization of Galinstan-based capacitive sensor. (a) optimization of wireless communication distance (0–10 mm) and (b) wireless communication test on three different media (air, water, and saline).

Finally, the applicability of Galinstan-based capacitive sensor for wireless human motion monitoring was demonstrated by fixing the capacitive sensor on an index finger and wrist, as shown in Fig. 7(a) and (d), respectively. The capacitive sensor mounted on the joint of the index finger caused various deformation at bending angles, from 0 to 90°. Deformation of the capacitor due to finger motion reduced the thickness of the dielectric medium. As a result, the capacitance value was increased and the resonant frequency of the sensor is lowered, as shown in the Fig. 7(b). Fig. 7(c) shows the change of capacitance over time for various bending angles of the index finger joint. The gradual change in resonant frequency and capacitance shows the influence of finger motion on the capacitive sensor. Similarly, the capacitive sensor attached on the wrist developed strain associated with wrist flexion and release for various angles from 0 to 60°. The corresponding change in resonant frequency of the sensor for different wrist positions is shown in Fig.7(e), while Fig.7(f) demonstrates the change in capacitance as a function of time. Thus, the real-time data obtained from the capacitive sensor induced by specific human-motion, confirms the possibility of the proposed sensor for wireless human-motion monitoring.

3.3. *Wireless sensing by readout method*

Wireless communication with Galinstan-based capacitive sensor was achieved via inductive coupling between a sensor antenna and an external readout coil. Here, the flexible sensor with a 58 MHz resonant frequency was utilized for the wireless communication test. As it can be observed in Fig. 8(a), the sensor communicated wirelessly up to 10 mm in air using the proposed readout method. The detection limit rapidly reduced as the distance increased, showing how the distance between the sensor coil and the external antenna was closely related to sensor sensitivity. Further, experiments were conducted on the electromagnetic field changes affected by the objects present between the two coils. In order to test the influence of the medium on inductive coupling, three media (i.e., air, water, and saline) were placed directly between the sensor and the readout coil. Corresponding results are shown in Fig. 8(b), indicating that the medium had a significant influence on wireless communications. The resonant frequency of the sensor changed from 55 MHz to 58 MHz according to the three media, owing to the high permittivity of water-based solutions. The water-based solutions have a dielectric constant of 80, whereas the air has 1. In general, the high permittivity increases the electric field concentration in the liquid medium, which effectively increases the equivalent capacitance and decreases the resultant phase-dip frequency. Thus, the quality factor of the liquid dropped drastically, compared to the air medium, decreasing the sensing distance of the liquid-based medium.

This work demonstrates the potential of gallium liquid metal alloy in flexible microfluidic devices for wireless sensor applications. Though, various microfluidic devices, including sensors, pumps, capacitive electrodes, and valves etc. have been reported [14], multilayer microfluidic device with wireless sensing platform is addressed less because of its complex structure, as the microchannels of different layers have to be connected in order to integrate all the components of the circuit. Based on this work, we believe that the use of microfluidic injection method in the fabrication process makes it easy to realize the multilayer microfluidic device, which is more difficult by 3D printing, selective plating or any other techniques. Thus the Galinstan based multilayer microfluidic device opens the option for realizing complex and new sensing techniques that demand flexibility, stretchability, and conformability, especially in biomedical applications and wireless flexible electronics.

4. Conclusion

We demonstrated a flexible microfluidic device-based wireless capacitive sensor, built using Galinstan for wireless sensor applications. The PDMS microchannel was treated with H_2SO_4, improving the non-wetting characteristics of Galinstan inside the microchannels. Electromechanical properties of the sensor were evaluated through repeated pressure loading and unloading cycles with the strain range of 0 %–30 %. Further, the wireless sensor was fixed on the wrist and index finger to measure human motion at various bending angles. The sensitivity of the flexible sensor measured at a pressure range of 0–200 mmHg was 5 kHz/mmHg, which was obtained by the wireless readout method. In the case of air medium, the flexible sensor could detect the pressure change in real time at a working distance of up to 10 mm. The sensitivity of the pressure sensor in the air medium was compared with water and saline medium. Finally, we hope that the proposed Galinstan-based flexible pressure sensors can be applied in various fields such as wireless human-machine interfaces and soft-skin electronic devices.

Authorship contributions

All persons who meet authorship criteria are listed as authors, and all authors certify that they have participated sufficiently in the work to take public responsibility for the content, including participation in the concept, design, analysis, writing, or revision of the

manuscript. Furthermore, each author certifies that this material or similar material has not been and will not be published in any other publication before its appearance in the Sensors and Actuators A: Physical.

We confirm that the manuscript has been read and approved by all the above named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

CRediT authorship contribution statement

Karthikeyan Munirathinam: Conceptualization, Methodology, Software, Investigation, Writing - original draft. **Jongsung Park:** Resources, Visualization, Data curation, Supervision. **Yun-Jin Jeong:** Software, Validation. **Dong-Weon Lee:** Project administration, Funding acquisition, Writing - review & editing.

Declaration of Competing Interest

None.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.sna.2020.112344.

References

[1] T. Yamada, Y. Hayamizu, Y. Yamamoto, Y. Yomogida, A. Izadi-Najafabadi, D.N. Futaba, K. Hata, A stretchable carbon nanotube strain sensor for human-motion detection, Nat. Nanotechnol. 6 (5) (2011) 296–301.
[2] S. Shengbo, L. Lihua, J. Aoqun, D. Qianqian, J. Jianlong, Z. Qiang, Z. Wendong, Highly sensitive wearable strain sensor based on silver nanowires and nanoparticles, Nat. Nanotechnol. 29 (25) (2018), 255202.
[3] J. Park, J.K. Kim, D.S. Kim, A. Shanmugasundaram, S.A. Park, S. Kang, S.H. Kim, M.H. Jeong, D.W. Lee, Wireless pressure sensor integrated with a 3D printed polymer stent for smart health monitoring, Sens. Actuators B Chem. 280 (2019) 201–209.
[4] Y.Q. Li, W.B. Zhu, X.G. Yu, P. Huang, S.Y. Fu, N. Hu, K. Liao, Multifunctional wearable device based on flexible and conductive carbon sponge/polydimethylsiloxane composite, ACS Appl. Mater. Interface 8 (48) (2016) 33189–33196.
[5] A.K.K. Kyaw, H.H.C. Loh, F. Yan, J. Xu, A polymer transistor array with a pressure-sensitive elastomer for electronic skin, J. Mater. Chem. C Mater. Opt. Electron. Devices 5 (46) (2017) 12039–12043.
[6] S. Zhang, L. Wen, H. Wang, K. Zhu, M. Zhang, Vertical CNT–Ecoflex nanofins for highly linear broad-range-detection wearable strain sensors, J. Mater. Chem. C 6 (19) (2018) 5132–5139.
[7] L. Nela, J. Tang, Q. Cao, G. Tulevski, S.J. Han, Large-area high-performance flexible pressure sensor with carbon nanotube active matrix for electronic skin, Nano Lett. 18 (3) (2018) 2054–2059.
[8] Y.H. Kwak, W. Kim, K.B. Park, K. Kim, S. Seo, Flexible heartbeat sensor for wearable device, Biosens. Bioelectron. 94 (2017) 250–255.
[9] X. Shuai, P. Zhu, W. Zeng, Y. Hu, X. Liang, Y. Zhang, R. Sun, C.P. Wong, Highly sensitive flexible pressure sensor based on silver nanowires-embedded polydimethylsiloxane electrode with microarray structure, ACS Appl. Mater. Interface 9 (31) (2017) 26314–26324.
[10] S. Han, J. Kim, S.M. Won, Y. Ma, D. Kang, Z. Xie, K.T. Lee, H.U. Chung, A. Banks, S. Min, S.Y. Heo, Battery-free, wireless sensors for full-body pressure and temperature mapping, Sci. Transl. Med. 10 (435) (2018) 4950.
[11] G. Bo, L. Ren, X. Xu, Y. Du, S. Dou, Recent progress on liquid metals and their applications, Adv Phys-x 3 (1) (2018), 1446359.
[12] B. Gol, F.J. Tovar-Lopez, M.E. Kurdzinski, S.Y. Tang, P. Petersen, A. Mitchell, K. Khoshmanesh, Continuous transfer of liquid metal droplets across a fluid–fluid interface within an integrated microfluidic chip, Lab Chip 15 (11) (2015) 2476–2485.
[13] M.D. Dickey, Emerging applications of liquid metals featuring surface oxides, ACS Appl. Mater. Interface 6 (21) (2014) 18369–18379.
[14] K. Khoshmanesh, S.Y. Tang, J.Y. Zhu, S. Schaefer, A. Mitchell, K. Kalantar-Zadeh, M.D. Dickey, Liquid metal enabled microfluidics, Lab Chip 17 (6) (2017) 974–993.
[15] D. Kim, P. Thissen, G. Viner, D.W. Lee, W. Choi, Y.J. Chabal, J.B. Lee, Recovery of nonwetting characteristics by surface modification of gallium-based liquid metal droplets using hydrochloric acid vapor, ACS Appl. Mater. Interface 5 (1) (2012) 179–185.
[16] G. Li, M. Parmar, D.W. Lee, An oxidized liquid metal-based microfluidic platform for tunable electronic device applications, Lab Chip 15 (3) (2015) 766–775.
[17] L.Y. Zhou, Q. Gao, J.F. Zhan, C.Q. Xie, J.Z. Fu, Y. He, Three-dimensional printed wearable sensors with liquid metals for detecting the pose of snakelike soft robots, ACS Appl. Mater. Interface 10 (27) (2018) 23208–23217.
[18] G. Li, X. Wu, D.W. Lee, A galinstan-based inkjet printing system for highly stretchable electronics with self-healing capability, Lab Chip 16 (8) (2016) 1366–1373.
[19] G. Li, D.W. Lee, An advanced selective liquid-metal plating technique for stretchable biosensor applications, Lab Chip 17 (20) (2017) 3415–3421.
[20] J. ChuanáYeo, C. TeckáLim, Emergence of microfluidic wearable technologies, Lab Chip 16 (21) (2016) 4082–4090.
[21] S. Cheng, Z. Wu, Microfluidic electronics, Lab Chip 12 (16) (2012) 2782–2791.
[22] M.D. Dickey, Stretchable and soft electronics using liquid metals, Adv. Mater. 29 (27) (2017), 1606425.
[23] J.C. Yeo, J. Yu, Z.M. Koh, Z. Wang, C.T. Lim, Wearable tactile sensor based on flexible microfluidics, Lab Chip 16 (17) (2016) 3244–3250.
[24] S.W. Jin, J. Park, S.Y. Hong, H. Park, Y.R. Jeong, J. Park, S. S.Lee, J.S. Ha, Stretchable loudspeaker using liquid metal microchannel, Sci.Rep. 5 (2015) 11695.
[25] Y. Gao, H. Ota, E.W. Schaler, K. Chen, A. Zhao, W. Gao, H.M. Fahad, Y. Leng, A. Zheng, F. Xiong, C. Zhang, Wearable microfluidic diaphragm pressure sensor for health and tactile touch monitoring, Adv. Mater. 29 (39) (2017), 1701985.
[26] S.G. Yoon, H.J. Koo, S.T. Chang, Highly stretchable and transparent microfluidic strain sensors for monitoring human body motions, ACS Appl. Mater. Interface 7 (49) (2015) 27562–27570.
[27] J. Yan, Y. Lu, G. Chen, M. Yang, Z. Gu, Advances in liquid metals for biomedical applications, Chem. Soc. Rev. 47 (8) (2018) 2518–2533.
[28] Y.R. Jeong, J. Kim, Z. Xie, Y. Xue, S.M. Won, G. Lee, S.W. Jin, S.Y. Hong, X. Feng, Y. Huang, J.A. Rogers, A skin-attachable, stretchable integrated system based on liquid GaInSn for wireless human motion monitoring with multi-site sensing capabilities, NPG Asia Mater. 9 (10) (2017) 443.
[29] M. Varga, C. Ladd, S. Ma, J. Holbery, G. Tröster, On-skin liquid metal inertial sensor, Lab Chip 17 (19) (2017) 3272–3278.
[30] P.J. Chen, S. Saati, R. Varma, M.S. Humayun, Y.C. Tai, Wireless intraocular pressure sensing using microfabricated minimally invasive flexible-coiled LC sensor implant, J. Microelectromech. Syst. 19 (4) (2010) 721–734.
[31] Q.A. Huang, L. Dong, L.F. Wang, LC passive wireless sensors toward a wireless sensing platform: status, prospects, and challenges, J. Microelectromech. Syst. 25 (5) (2016) 822–841.
[32] H.H. Hsieh, F.C. Hsu, Y.F. Chen, Energetically autonomous, wearable, and multifunctional sensor, ACS Sens. 3 (1) (2018) 113–120.
[33] J. Heikenfeld, A. Jajack, J. Rogers, P. Gutruf, L. Tian, T. Pan, R. Li, M. Khine, J. Kim, J. Wang, Wearable sensors: modalities, challenges, and prospects, Lab Chip 18 (2) (2018) 217–248.
[34] Q. Gao, H. Li, J. Zhang, Z. Xie, J. Zhang, L. Wang, Microchannel structural design for a room-temperature liquid metal based super-stretchable sensor, Sci. Rep. 9 (1) (2019) 5908.
[35] L.Y. Chen, B.C.K. Tee, A.L. Chortos, G. Schwartz, V. Tse, D.J. Lipomi, H.S.P. Wong, M.V. McConnell, Z. Bao, Continuous wireless pressure monitoring and mapping with ultra-small passive sensors for health monitoring and critical care, Nat. Commun. 5 (1) (2014) 5028.
[36] D. Kwon, T.I. Lee, J. Shim, S. Ryu, M.S. Kim, S. Kim, T.S. Kim, I. Park, Highly sensitive, flexible, and wearable pressure sensor based on a giant piezo capacitive effect of three-dimensional microporous elastomeric dielectric layer, ACS Appl. Mater. Interface 8 (26) (2015) 16922–16931.
[37] S. Mutashar, M.A. Hannan, S.A. Samad, A. Hussain, Analysis and optimization of spiral circular inductive coupling link for bio-implanted applications on air and within human tissue, Sensors 14 (7) (2014) 11522–11541.
[38] N. Lazarus, C.D. Meyer, W.J. Turner, A microfluidic wireless power system, RSC Adv. 5 (96) (2015) 78695–78700.
[39] N. Xue, S.P. Chang, J.B. Lee, A SU-8-based microfabricated implantable inductively coupled passive RF wireless intraocular pressure sensor, J. Microelectromech. Syst. 21 (6) (2012) 1338–1346.
[40] J. Kim, A. Banks, Z. Xie, S.Y. Heo, P. Gutruf, J.W. Lee, S. Xu, K.I. Jang, F. Liu, G. Brown, J. Choi, J.H. Kim, X. Feng, Y. Huang, U. Paik, J.A. Rogers, Miniaturized flexible electronic systems with wireless power and near-field communication capabilities, Adv. Funct. Mater. 25 (30) (2015) 4761–4767.

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